INCH-POUND
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DETAIL SPECIFICATION

CLOTH, CAMOUFLAGE PATTERN: COTTON AND NYLON

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers the following classes as specified (see 6.2).

1.2 Classification. The cloth is furnished in the following classes as specified (see 6.2).

Class 1 - Camouflage Printed (Woodland Camouflage Print)
Class 2 - Camouflage Printed and Quarpel Treated (Woodland Camouflage Print)
Class 3 - Universal Camouflage Print (UCP), Static Dissipative Fiber
Class 4 - Universal Camouflage Print (UCP), Water Repellent Treated
Class 5 - Woodland Camouflage Print, Static Dissipative Fiber
Class 6 - Universal Camouflage Print (UCP)
Class 7 - Operation Enduring Freedom Camouflage Pattern (OCP) (see 6.10)
Class 8 - Operation Enduring Freedom Camouflage Pattern (OCP), Water Repellent Treated (see 6.10)

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: Attn: DLA Troop Support Standardization Team, 700 Robbins Avenue, Philadelphia, PA 19111-5096. Since contact information can change, you may want to verify the currency of the address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database https://assist.daps.dla.mil/.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-4 - Glossary of Fabric Imperfections

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-50199 - Thread, Polyester Core, Cotton or Polyester-Covered

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or from the Standardization Order Desk, Building 4D, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

DRAWINGS

2-1-1516 - Woodland Camouflage Pattern
2-1-2519 - Universal Camouflage Pattern

(Copies of drawings are available from the Department of the Army, Natick Soldier Research Development and Engineering Center, Kansas St., Natick MA 01760, ATTN: RDNS-WPW-C)

FEDERAL TRADE COMMISSION

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies are available online at http://www.ftc.gov or from the Federal Trade Commission, 600 Pennsylvania Avenue, N.W., Washington, DC 20580-0001.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC Test Method 8 - Colorfastness to Crocking: AATCC Crockmeter Method
AATCC Test Method 15 - Colorfastness to Perspiration
AATCC Test Method 16 - Colorfastness to Light
AATCC Test Method 20 - Fiber Analysis: Qualitative
AATCC Test Method 20A - Fiber Analysis: Quantitative
AATCC Test Method 22 - Water Repellency – Spray Test
AATCC Test Method 61 - Colorfastness to Laundering: Accelerated
AATCC Test Method 70 - Water Repellency: Tumble jar Dynamic Absorption Test
AATCC Test Method 81 - pH of the Water-Extract from Wet Processed Textiles
AATCC Test Method 96 - Dimensional Changes in Commercial Laundering of Woven and Knitted Fabrics Except Wool
AATCC Test Method 118 - Oil Repellency: Hydrocarbon Resistance Test
AATCC Test Method 127 - Water Resistance: Hydrostatic Pressure Test
AATCC Evaluation Procedure 1 - Gray Scale for Color Change
AATCC Evaluation Procedure 2 - Gray Scale for Staining
AATCC Evaluation Procedure 8 - AATCC 9-Step Chromatic Transference Scale
AATCC Evaluation Procedure 9 - Visual Assessment of Color Difference of Textiles

(Copies of these documents are available online at http://www.aatcc.org or from the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709-2215.)

AMERICAN SOCIETY FOR QUALITY

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies are available online at http://www.asq.org or from the American Society for Quality Control, 600 North Plankinton Avenue, Milwaukee, WI 53203.)

ASTM INTERNATIONAL

ASTM D 276 - Standard Test Methods for Identification of Fibers in Textiles
ASTM D 629 - Standard Test Methods for Quantitative Analysis of Textiles
ASTM D 737 - Standard Test Method for Air Permeability of Textile Fabrics
ASTM D 1424 - Standard Test Method for Tearing Strength of Fabrics by Falling-Pendulum (Elmendorf) Apparatus
ASTM D 1683 - Standard Test Method for Failure in Sewn Seams of Woven Fabrics
ASTM D 3775 - Standard Test Method for Warp (End) and Filling (Pick) Count of Woven Fabrics
ASTM D 3776 - Standard Test Method for Mass per Unit Area (Weight) of Fabric
ASTM D 5034 - Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)

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(Copies of these documents are available on line at http://www.astm.org or from the ASTM INTERNATIONAL, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959.)

OTHER PUBLICATIONS

Principles and Methods of Toxicology, Repeat Insult Patch Test - Modified Draize Procedure - (fourth edition) A Wallace Hayes (editor), pp 1057 – 1060, 2001.

(Copies are available online at http://www.taylorandfrancis.co.uk/ or from Taylor and Francis, 325 Chestnut Street, Philadelphia PA 19106 .)

Sears Roebuck and Co.
Sears Fabric Defect Replica Scales
(Copies available from Sears Roebuck and Co., "Fabric Defect Replica Kit" Dept 817 (ATTN: FC 554B), 3333 Beverly Road, HG, FC568B, Hoffman Estates, IL 60179). For information call (847-286-8952.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Standard sample. The finished cloth shall match the standard sample for shade and appearance and shall, unless otherwise indicated, be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

3.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Materials.

3.4.1 Cotton. The cotton shall be carded.

3.4.2 Nylon. The nylon shall be first quality, high tenacity, semi-dull staple having a nominal cut length of 1-1/2 inches and a round cross section with a nominal denier of 2.25 to 2.50. No form of nylon waste shall be used, such as undrawn fiber, mixtures of deniers, lusters or cross sections, and waste from any stage of fiber production: whether drawn, un-drawn, or mixed or garneted fiber. Testing shall be as specified in 4.4.3.

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3.4.3 Static dissipative fiber (Class 3 and Class 5). The fiber shall have a carbon core and nylon sheath and shall be 3 to 3.8 nominal denier per filament and a cut length of 1-1/2 to 2 inches in staple length (see 6.8).

3.4.4 Yarns. The warp and filling yarn shall be singles made from a blend of 50 (± 5) percent nylon and the remaining percentage cotton based on the dry weight of the desized cloth when tested as specified in 4.4.3. The yarn for the Class 3 and Class 5 shall contain 2 percent static dissipative fiber.

3.5 Color.

3.5.1 Classes 1, 2 and 5 Woodland Camouflage. The cloth shall be dyed to a ground shade either matching or approximating Light Green 354 and then shall be overprinted with the camouflage pattern by roller or screen printing. When the ground shade is dyed to match Light Green 354, the remaining colors shall be obtained by subsequent printing using three rollers or screens, as appropriate for the Dark Green 355, Brown 356, and Black 357 areas of the pattern. When the ground shade is dyed to approximate Light Green 354, all four colors of the camouflage pattern shall be obtained by subsequent printing using four rollers or screens to match all four colors. Resin bonded pigments shall not be used (see 6.5).

3.5.2 Classes 3, 4 and 6 Universal Camouflage Pattern (UCP). The cloth shall be dyed to a ground shade either matching or approximating Desert Sand 500 and then shall be overprinted with the camouflage pattern by roller or screen printing. When the ground shade is dyed to match Desert Sand 500, the remaining colors shall be obtained by subsequent printing using two rollers or screens, as appropriate for the Urban Gray 501 and Foliage Green 502 areas of the pattern. When the ground shade is dyed to approximate Desert Sand 500 all three colors of the camouflage pattern shall be obtained by subsequent printing using three rollers or screens to match all three colors. Resin bonded pigments shall not be used (see 6.5).

3.5.3 Classes 7 and 8, Operation Enduring Freedom Camouflage Pattern (OCP). The cloth shall be dyed to a ground shade either matching or approximating Cream 524 and then shall be overprinted with the camouflage pattern by roller or screen printing. When the ground shade is dyed to match Cream 524, the remaining colors shall be obtained by subsequent printing using six rollers or screens, as appropriate for the Tan 525, Pale Green 526, Olive 527, Dark Green 528, Brown 529 and Dark Brown 530 areas of the pattern. When the ground shade is dyed to approximate Cream 524 all seven colors of the camouflage pattern shall be obtained by subsequent printing using seven rollers or screens to match all seven colors. Resin bonded pigments shall not be used (see 6.5).

3.5.3.1 Labile sulfur. Dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid shall not be used. The results shall be free (no discoloration) when tested in accordance with 4.4.3.

3.5.4 Visual shade matching (all Classes). The color and appearance of the camouflage printed cloth shall match the standard sample when tested as specified in 4.4.3.

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3.5.5 Colorfastness (all Classes). The finished cloth shall conform to the colorfastness requirements listed below in Tables I and IA when tested as specified in 4.4.3 and 4.5.1.

TABLE I. Colorfastness requirements (Classes 1 - 6).

Colors Evaluation	Laundering (3 cycles) 1/ (minimum)	Light (40 hrs Or 170 kJ) 2/ (minimum)	Perspiration (acid & alkaline) 1/ (minimum)	Crocking 3/ (minimum)
Black 357	3	3-4	3-4	1
Dk. Green 355, Brown 356, Lt. Green 354, Desert Sand 500, Urban Gray 501, Foliage Green 502	3-4	3	3-4	3.5

1/ Rated using the AATCC Evaluation Procedure 1, Gray Scale for Color Change and AATCC Evaluation Procedure 2, Gray Scale for Staining.
2/ Rated using the AATCC Evaluation Procedure 1, Gray Scale for Color Change
3/ Rated using the AATCC Evaluation Procedure 8, AATCC 9-Step Chromatic Transference Scale.

TABLE IA. Colorfastness requirements (Classes 7 and 8).

Colors Evaluation	Laundering (3 cycles) 1/ (minimum)	Light (40 hrs or 170 kJ) 2/ (minimum)	Perspiration (acid & alkaline) 1/ (minimum)	Crocking 3/ (minimum)
All colors	3-4	----	3-4	3.5
Dk. Green 528, Brown 529, Dark Brown 530	----	3-4	----	----
Cream 524, Tan 525, Pale Green 526, Olive 527	----	3	----	----

1/ Rated using the AATCC Gray Scale for Color Change and AATCC Gray Scale for Staining.
2/ Rated using the AATCC Gray Scale for Color Change.
3/ Rated using the AATCC 9-Step Chromatic Transference Scale.

3.5.6 Pattern execution. The pattern on the printed finished cloth(s) shall reproduce the standard sample in respect to design, colors and registration of the respective areas. The pattern repeat of Class 1, 2 and 5 shall be 27.25 (+1.25, -2.50) inches in the warp direction. The pattern repeat of Class 3, 4 and 6 shall be 36.00 (+1.25, -2.50) inches in the warp direction. The pattern repeat of Class 7 and 8 shall be 25.255 (+1.25, -2.50) inches in the warp direction. The various areas of the pattern shall be properly registered in relation to each other and shall present definite sharp demarcations with a minimum of feathering or spew. Each pattern area shall show solid

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coverage; skitteriness exceeding that shown on the standard sample in any of the printed areas will not be acceptable. When the standard sample is not referenced for pattern execution, a pattern drawing will be provided and the pattern for Class 1, 2 and 5 shall match that of Drawing 2-1-1516, and for Class 3, 4 and 6 shall match that of Drawing 2-1-2519 (see 2.2.2, 6.2, 6.4 and 6.7).

3.5.7 Spectral reflectance Classes 1, 2 and 5. The spectral reflectance values for each color in the Woodland Camouflage printed cloth shall conform to the requirements specified in Table II, when tested as specified in 4.4.3.

TABLE II. Spectral reflectance requirements, Classes 1, 2 and 5

Reflectance values (percent) - Woodland Camouflage						
Wavelength Nanometers (nm)	Black 357		Light Green 354		Dark Green 355 and Brown 356	
	min.	max.	min.	max.	min.	max.
600	-	10	8	18	3	9
620	-	10	8	18	3	9
640	-	10	8	18	3	9
660	-	10	8	18	3	12
680	-	10	10	22	3	14
700	-	10	18	33	5	18
720	-	10	22	45	7	20
740	-	10	30	55	12	28
760	-	10	35	65	18	36
780	-	10	40	75	26	44
800	-	10	45	80	34	52
820	-	10	50	86	42	60
840	-	10	55	88	50	68
860	-	10	60	90	56	74

3.5.8 Spectral reflectance Classes 3, 4 and 6. The spectral reflectance values for each color in the Universal Camouflage Pattern printed cloth shall conform to the requirements specified in Table III, when tested as specified in 4.4.3.

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TABLE III. Spectral reflectance requirements, Class 3, 4 and 6.

Reflectance values (percent) – Universal Camouflage Pattern (UCP)						
Wavelength Nanometers (nm)	Desert Sand 500		Urban Gray 501		Foliage Green 502	
	min.	max.	min.	max.	min.	max.
600	28	40	12	26	8	18
620	30	42	14	26	8	18
640	34	48	14	28	8	20
660	38	56	14	30	10	26
680	44	60	18	34	10	26
700	46	66	24	38	12	28
720	48	68	26	42	16	30
740	48	72	30	46	16	30
760	50	74	32	48	18	32
780	54	76	34	48	18	34
800	54	76	34	50	20	36
820	54	76	36	54	22	38
840	56	78	38	54	24	40
860	56	78	40	56	26	42

3.5.9 Spectral reflectance, Classes 7 and 8. The spectral reflectance values for each color in the Operation Enduring Freedom Camouflage Pattern cloth shall conform to the requirements specified in Table IIIA, when tested as specified in 4.4.3.

TABLE IIIA. Spectral reflectance requirements, Class 7 and 8.

Reflectance values (percent) – Operation Enduring Freedom Camouflage Pattern (OCP)						
Wavelength, Nanometers (nm)	Cream 524 & Tan 525		Pale Green 526, Olive 527 and Brown 529		Dark Green 528 and Dark Brown 530	
	min.	max.	min.	max.	min.	max.
600	22	44	12	30	3	11
620	24	45	12	30	3	11
640	24	45	12	32	4	12
660	25	45	12	32	4	12
680	28	45	14	34	4	13
700	28	46	14	34	6	16
720	30	48	16	36	6	20
740	32	50	18	36	10	25
760	36	50	20	40	14	30
780	38	52	22	40	18	35
800	40	54	22	42	22	40
820	44	56	24	44	24	42
840	46	57	26	44	27	43
860	48	58	28	46	29	45

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3.6 Physical requirements. The finished cloth shall conform to the requirements in Table IV when tested as specified in 4.4.3.

TABLE IV. Physical requirements

Characteristic	Classes 1, 3, 5, 6 and 7	Classes 2, 4 and 8
Weight (oz./sq.yd.), min.	6.8	7.0
Construction, (yarns per inch), min.		
Warp	86	86
Filling	54	54
Breaking strength (pounds), min.		
Warp	200	190
Filling	125	115
Tearing strength (pounds), min.		
Warp	11	10
Filling	8	7
Air permeability (cu.ft/minute/sq. ft), min.	25.0	10.0
Electrostatic Decay (Class 3 and Class 5) Initial and after five launderings	See 3.10	See 3.10

3.6.1 Weave. The weave shall be a 2/1 left hand twill.

3.6.2 Width. For Government procurements only, the width of the cloth shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of the selvage when fly-shuttle looms or shuttleless with tuck-in selvage are used. For all other shuttleless looms, the width measurement shall be made between the last warp yarn on each side excluding the protruding fringe(s).

3.7 Finish. The cloth shall be dyed and printed with the warp effect side as the face. The cloth shall be, desized, scoured, dyed, and printed. The Class 2, 4 and 8 cloths shall be given a water repellent treatment as specified in 3.7.1.

3.7.1 Water repellency (Class 2, 4 and 8). The cloth shall be given a fluorocarbon (Quarpel Type) water repellent treatment meeting the requirements specified in Table V and 3.9. Testing shall be as specified in 4.4.3.

TABLE V. Water repellency (Class 2, 4 and 8)

	Dynamic absorption (percent) max. 1/	Hydrostatic height (centimeters) min.	Spray rating (min)
Initial	20	18.0	100, 100, 90
After 15 launderings	20	18.0	----

1/ No individual specimen shall exceed the specified maximum.

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3.8 pH. The pH value of the water extract of the finished cloth shall be not less than 5.0 and not greater than 8.5 when tested as specified in 4.4.3.

3.9 Liquid repellency (Class 2, 4 and 8). The treated finished Class 2, 4 and 8 cloth shall show no wetting by n-dodecane after 30 seconds and no wetting by tri-ethyl phosphate, di-methyl phosphonate, and bis (2ethyl hexyl) hydrogen phosphate after 8 hours when tested as specified in 4.4.3.

3.10 Electrostatic decay (Class 3 and Class 5). The finished Class 3 and Class 5 cloths when tested as specified in 4.4.3 shall dissipate 90 percent of the charge in less than or equal to 1/2 second. The average warp results and average filling results shall meet this requirement before and after five launderings when tested as specified in 4.4.3.

3.11 Dimensional stability. The shrinkage or elongation both in the warp and in the filling of the finished cloth shall be not greater than 3.5 percent for the individual sample unit and not greater than 3.0 percent for the lot average when tested as specified in 4.4.3. The preshrinkage process used shall not be identified by name or trademark on the cloth, ticket, or package.

3.12 Seam efficiency. The finished cloth shall have a seam efficiency of no less than 80 percent when tested as specified in 4.4.3.

3.13 Toxicity. The finished cloth shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.4.3. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used.

3.14 Length and put-up. For Government procurements only, unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards. Each length shall be put-up on a roll as specified (see 6.2).

3.15 Fiber identification. Each roll shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.16 Workmanship. The finished cloth shall conform to the quality of product established by this specification. The demerit points per 100 square yards when calculated as specified in Section 4 shall not exceed the applicable established maximum point values.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. A first article, submitted in accordance with 3.1, shall be inspected, examined for appearance, color and finished defects listed in Table VII and tested for

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the characteristics as specified in Table VIII. Unless otherwise specified in the contract, the sample unit shall be one continuous 5 yard length of full width, finished cloth (see 6.3).

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.4 and the tests of 4.4.2 through 4.5.5 as applicable.

4.3.1 Inspection conditions. In accordance with 4.1, all inspections shall be performed in accordance with all the requirements of referenced documents, unless otherwise excluded, amended, modified or qualified in this specification or applicable procurement documents. (See 6.2)

4.3.2 Conformance inspection samples. Sampling for finished cloth testing shall be performed in accordance with Table VI. Unless otherwise specified in the contract, the sample unit shall be 5 continuous yards full width of the finished cloth per roll of cloth.

TABLE VI. Lot sample size

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

4.4 Examination. Each roll in the sample shall be examined yard-by-yard on the printed side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. The sample size shall be 20 rolls selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards examined exceeds 20.0 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceeds 30.0 points. If one roll in the sample exceeds 30.0 points per 100 square yards, a second sample of 20 rolls shall be examined for individual roll quality only. The lot shall be unacceptable if one or more rolls in the second sample exceed 30.0 points per 100 square yards. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{Total yards inspected}} = \text{Points per 100 square yards}$$

All defects as defined in section III of FED-STD-4 that are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.4.1

Except as follows:

a. Only coarse yarns that exceed twice the normal diameter shall be scored.
b. Mixed filling (shade bar shall be scored only when resulting from wrong ply, wrong twist in the yarn, or off shade yarn.
c. Only knots and slubs that exceed limits shown on Sears Fabric Defect Scale (see 2.3) “F” for slubs and “D” for knots shall be scored.

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The defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. No linear yard (increments of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points.

4.4.1 Demerit points. Demerit points shall be assigned as follows:

For defects up to 3 inches in any dimension	-	one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	-	two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	-	three points
For defects exceeding 9 inches in any dimension	-	four points

All defects as specified in Table VII shall be scored four points for each yard in which they occur:

TABLE VII. Defects

Defect Description
Any hole, cut, pinhole, tear, scratch or abrasion mark 1/
Edge ravels when pulled outward
Uneven weaving throughout
Pattern design not equal to the standard sample
Incorrect color in any part of the pattern
Pattern repeat not equal to the standard sample
Pattern repeat less than 24.75 inches or more than 28.50 inches (Classes 1, 2 and 5)
Pattern repeat less than 33.50 inches or more than 37.25 inches (Classes 3, 4 and 6)
Pattern repeat less than 22.755 inches or more than 26.505 inches (Classes 7 and 8)
Skitteriness (mottled, uneven color) of pattern exceeds that shown by the standard sample
Excessive feathering or spew (fuzziness at color boundaries) of pattern as compared to the standard sample
Excessive grinning (off register, gap where ground shade shows through) of pattern as compared to the standard sample
Excessive haloing or trapping (overlapping of colors) of pattern as compared to the standard sample
Overall application of water repellent not uniform (Classes 2 and 4)
Tackiness (sticky to touch) (Classes 2 and 4)
Design not printed on face side of cloth
Any spot, stain or streak more than 1 inch in combined directions. 2/
Any objectionable odor 3/
Width not as specified within a tolerance of ± 1/2 inch.
Broken or missing yarn. 2/

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TABLE VII. Defects - Continued

Defect Description
Any embedded foreign matter or uncleanness. 2/
Color not as specified
Slack or tight selvages 4/
Edges rolled folded, scalloped or corded, baggy, ridgy or wavy cloth.

1/ The same yardage shall be given a through –lighting inspection for pinholes and thinly coated areas.
2/ Clearly visible at normal inspection distance – (3 feet) by either direct viewing or through-lighting.
3/ Odors of chemicals commonly used in coating compounds shall not be considered objectionable.
4/ To determine the presence of unacceptable selvage conditions, the following procedure shall be observed: During the visual examination, the perch shall be stopped a minimum of three times for each roll in the sample, the tension removed, and the finished cloth examined for the selvage conditions. A waviness in the selvage or significant ripples diagonally across the width of the fabric is an indication of slack or tight selvages.

4.4.1.1 Length examination. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets.

4.4.2 Shade and appearance examination. The shade and appearance examination sampling procedure shall be followed as written below.

A) Sampling: a 12 inch by full width swatch (header) will be cut by the contractor from each piece or roll in the lot. The header will be cut in half to form Set A and Set B. The Government representative will select samples in accordance with the following:

Lot size (pieces)	pieces to be sampled
1 to 5	each piece
6 to 25	5
26 to 90	10
91 to 160	16
over 160 pieces	1 of every 10 pieces

B) The swatches shall be identified and submitted to the DLA Troop Support Laboratory for shade evaluation. If one or more of the shade swatches are found unacceptable, the entire lot shall be rejected. Any end items made from the component lot shall also be rejected. A lot that was rejected for shade shall, without cost to the Government, be screened and all defective pieces in the lot shall be removed before such a lot is resubmitted. Resubmitted lots shall again be subjected to the sampling and shade evaluation prescribed herein. This requirement does not negate the contractor's

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responsibility to perform shade evaluation prior to the submittal of a lot to the Government.
.
(c) When Section 3 of the fabric specification contains a specific requirement for uniformity of shade, the swatches submitted in accordance with (a) above shall also be evaluated for uniformity of shade.
.
(d) If the contractor reworks and resubmits pieces originally rejected for shade or finish, the contractor shall not combine the rejected pieces with normal production or with lots rejected for other causes. Such pieces shall be combined to form one resubmitted lot. Each piece shall retain it original piece number, suffixed with an "x". The lot number shall also be suffixed with an "x'.
.
(e) Shade sampling procedures are detailed further in DLA Troop Support Manual, Quality Requirements, PSCM 4155.3, Part 1, which is incorporated by reference.
The swatches shall be submitted to the following:

Commander
DLA Testing Center
Naval Support Station Philadelphia
700 Robbins Ave (Building 5 D)
Philadelphia. Pa. 19111-5098

4.4.3 End item testing. The cloth shall be tested for the characteristics listed in Table VIII. The methods of testing as specified wherever applicable and as listed in Table VIII shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The lot shall be unacceptable if one or more sample units or the lot average for dimensional stability fail to meet any requirement specified.

TABLE VIII. End item tests

Characteristic	Requirement reference	Test method
Fiber Identification	3.4.1-3.4.3	AATCC-20 or ASTM-D-276 1/
Fiber content %	3.4.4	AATCC-20A or ASTM-D-629 1/ 2/
Yarn Ply	3.4.4	Visual
Labile sulfur	3.5.3.1	4.5.1
Visual shade matching	3.5.4	AATCC Evaluation Procedure 9, Option A, 3/
Colorfastness: Laundering (after 3 cycles) Light (after 40 hrs or 170.kilojoules) Perspiration (acid & alkaline) Crocking	3.5.5 Table I Table I Table I Table I	(see 4.5.2) AATCC 61 Test 3A 4/ AATCC 16 Opt 1 or 3 AATCC 15 AATCC 8
Pattern execution	3.5.6	Visual

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TABLE VIII. End item tests – Continued

Characteristic	Requirement reference	Test method
Spectral reflectance:		
Classes 1, 2 and 5	3.5.7, Table II	4.5.3
Classes 3, 4 and 6	3.5.8, Table III	4.5.3
Classes 7 and 8	3.5.9, Table IIIA	4.5.3
Weight	3.6, Table IV	ASTM D 3776
Yarns per Inch	3.6, Table IV	ASTM D 3775
Breaking Strength	3.6, Table IV	ASTM D 5034
Tearing strength	3.6, Table IV	ASTM D 1424
Air Permeability	3.6, Table IV	ASTM D 737
Weave	3.6.1	Visual
Water repellent treatment 4/ (Classes 2, 4 and 8)	3.7	
Dynamic absorption		
Initial	Table V	AATCC 70
After 15 Launderings	Table V	AATCC 96 VIc, A 5/, 6/ and AATCC 70
Hydrostatic Pressure		
Initial	Table V	AATCC 127
After 15 Launderings	Table V	AATCC 96 VIc, A 5/, 6/ and AATCC 127
Spray rating:		
Initial	Table V	AATCC 22
pH	3.8	AATCC 81
Liquid repellency (Classes 2,4 and 8):		
Initial	3.9	AATCC 118
After 15 launderings	3.9	AATCC 96 VIc, A 5/, 6/ and AATCC 118 & 4.5.5
Electrostatic Decay (Classes 3 and 5)		
Initial	3.10	See 4.5.4
After 5 launderings	3.10	AATCC 96, VIc, A & 4.5.4 7/
Dimensional stability (after 1 cycle)	3.11	AATCC 96, VIc, A
Seam efficiency	3.12	ASTM 1683 8/
Toxicity	3.13	See 4.5.6

1/ In case of dispute, the ASTM method prevails.

2/ The cotton content shall be calculated as follows:
Cotton content, percent = R/S x 100
R = Weight of residual fibers
S = Weight of dry desized specimen

3/ The color and appearance of the camouflage printed cloth shall match the standard sample when viewed using the AATCC Evaluation Procedure 9, Option A, with sources simulating artificial daylight D75 illuminant with a color temperature of 7500 (+ 200)°K illumination of 100 (+ 20) foot candles, and shall be a good match to the standard sample under incandescent

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lamplight at 2856 (±200)°K.

4/ Only the stain on the nylon and cotton fibers of the color transfer cloth shall be evaluated.

5/ Specimens shall be subjected to 15 complete cycles (wash and dry) prior to determinations of dynamic absorption and resistance to organic liquid after laundering. The last two (2) wash cycles shall be performed without the addition of detergent.

6/ Immediately after tumble drying the specimens shall be laid out, without tension or pressing, on a flat surface in standard atmosphere until moisture equilibrium is reached.

7/ The last wash cycle shall be performed without the use of detergent. Specimens shall be dried after each laundering.

8/ The needle shall measure 0.040 (± 0.001) inch across the blade at the eye. The thread for all tapes shall be cotton or polyester covered in accordance with A-A-50199 Tex size 36-45, for the needle and top thread and Tex size 31-35 for the looper.

4.5 Methods of inspection.

4.5.1 Presence of labile sulfur test. In the determination of presence of labile sulfur in textile materials with lead acetate, two 1.50 (± 0.01) gram samples from each material submitted for evaluation shall be tested. Each of the two samples shall be cut into very small pieces and placed into separate test tubes. The samples shall be submersed in a stannous chloride solution that contains 100 grams of stannous chloride crystals ACS in 100 milliliters of hydrochloric acid ACS (35 percent concentration) and 50 milliliters of distilled water. A filter paper wet out with a 5.0 percent lead acetate solution shall be placed over the top of the test tube. The lead acetate solution contains 5.0 grams of lead acetate CP reagent grade and enough distilled water to make up a 100 milliliter solution; if the solution is not clear add a few drops (one at a time) of glacial acetic acid until the solution is clear. The test tube containing the textile sample, stannous chloride and wet filter paper shall be heated over a low flame until the solution is boiling. The solution should not be heated for more than 15 seconds. A brown to black stain on the filter paper should be evaluated as follows:

Free - The filter paper shows no discoloration or staining of any kind.
Slight - The filter paper shows a light tan to light brown discoloration stain.
Moderate - The filter paper shows a dark brown discoloration stain.
Severe - The filter paper shows a black color stain.

4.5.2 Colorfastness evaluation testing. When testing for colorfastness properties, each color shall be evaluated, whenever possible, separately and reported as such. In cases where the print pattern does not allow for the evaluation of each color separately, the test results should indicated which colors were evaluated together. All color evaluations to include visual, and colorfastness for Classes 7 and 8 shall be performed on the solid color area and not the tonal area.

4.5.3 Spectral reflectance test. Spectral reflectance data shall be obtained from 600 to 860 nanometers (nm) at 20 nm intervals on a spectrophotometer relative to the barium sulfate standard, the preferred white standard. Other white reference materials may be used provided they are calibrated to absolute white, e.g. magnesium oxide or vitrolite tiles. The spectral band width shall be less than 26 nm at 860 nm. Reflectance measurements shall be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode of operation is used, the spectrophotometer shall operate with the specimen diffusely illuminated with the full

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emission of a continuous source that simulates either CIE Source A or CIE Source D65. Measurements shall be taken on a minimum of two (2) different areas and the data averaged. The measured areas should be at least 6 inches away from the selvage. The specimen shall be measured as a single layer backed with layers of the same fabric and shade. For the Woodland pattern cloth, three backing layers shall be used for the Light Green 354, Dark Green 355 and Brown 356 colors and two backing layers for Black 357. All other camouflage pattern cloths shall be measured as a single layer backed with four layers of the same shade. The specimen shall be viewed at an angle no greater than 10° from normal, with the specular component included. Specimens shall be oriented in different directions during testing. When possible, the specimen tested shall not contain the same warp or filling yarns. Photometric accuracy of the spectrophotometer shall be within 1 percent and wavelength accuracy within 2 nm. The diameter for standard aperture size used in the color measurement device shall be 1.0 to 1.25 inches for Woodland camouflage and 0.3725 inches or larger for all other camouflage. Any color having spectral reflectance values falling outside the limits at four or more of the wavelengths specified shall be considered a test failure. The spectral reflectance for Classes 7 and 8 shall be performed on the solid color area and not the tonal area.

4.5.4 Electrostatic decay test. The specimen shall be tested according to the method specified in 4.5.4.1 through 4.5.4.5.

4.5.4.1 Apparatus. The apparatus shall consist of a humidity test chamber, a meter for measuring relative humidity and temperature, a dehumidifier, an air-ionizer or static eliminator, a faraday cage, a source capable of out putting 5000 volts, and voltage meters to display the applied voltage and the voltage detected by the sensor. The faraday cage shall contain two parallel electrodes, on which specimens may be mounted and charged, and a sensor that will detect voltage on the specimen surface. A staticmaster is needed to eliminate residual charges of 500 volts or less from the samples. A humidifier and a humidity controller may be added to the system to facilitate conditioning of the sample.

4.5.4.2 Test specimen. The test specimen shall be 3-inch by 6-inch rectangular piece of material. The warp or filling yarns shall run parallel to the 6-inch length of the specimen. The direction of test (warp or filling) shall be along the length of the specimen and each specimen shall be labeled accordingly in one corner. Test specimens shall be cut so that no two contain the same set of warp and filling yarns.

4.5.4.3 Number of determinations. Unless otherwise specified in the procurement document, six (6) specimens shall be tested, three (3) in the warp direction and three (3) in the filling direction, each at both sides: face and back.

4.5.4.4 Procedure. The faraday cage and specimens to be tested shall be preconditioned at 10 (± 2) percent relative humidity overnight, and conditioned at 20 (±2) percent relative humidity for a minimum of 24 hours at 75 (± 5) °F. The specimens shall be tested at 20 (± 2) percent relative humidity and 75 (± 5) °F. An air-ionizing blower or a static eliminator must be used in the chamber while conditioning but must be turned off once testing begins.

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4.5.4.4.1 Allow test equipment one-hour warm-up time. Adjust voltage source for a charging voltage of 5000 volts. Mount a 3-inch by 5-inch aluminum calibration plate across the electrodes in the faraday cage such that it is centered in front of the voltage sensor opening. Apply a charge of 5000 volts, and calibrate the voltage meter designated to indicate detected voltage on the sample equal to 5000 volts. Remove the aluminum plate.

4.5.4.4.2 Mount a specimen tautly across the electrodes, centering it over the sensor opening. The surface of test specimen (back or front) shall face the sensor. The CUT-OFF must be set to 10 percent. With the high voltage off, depress the CHG button to measure initial charge on the test specimen. If it is not reading zero, it may not be antistatic. A staticmaster shall be used on the specimen before the test to remove residual charge of 500 or less.

4.5.4.4.3 Apply 5000 volts to the electrodes for a period of 20 seconds. To do this, depress the + button on the right (the reading on the CHARGING VOLTAGE meter must read 5000 volts); then depress the CHG button. Wait 20 seconds; record the highest reading on the SAMPLE CHARGE meter. Then depress the TEST button and from the readout under DECAY TIME-SEC, record how long it takes for the charge to decay from the maximum voltage to 10 percent of the maximum level. Depress the ZERO/STBY button after the test. The high voltage (5000 volts) shall be turned off and the specimen immediately grounded.

4.5.4.4.4 If the maximum voltage level on the SAMPLE CHARGE meter did not reach at least 4000 volts during the 20-second charging period, the specimen shall be recorded as non-passing and the reason noted.

4.5.4.4.5 If the DECAY TIME-SEC is less than 0.5 second, the specimen is acceptable; otherwise it is not and the reason for failure shall be noted.

4.5.4.4.6 After testing the other side of the same specimen, remove it and repeat 4.5.4.4.2 through 4.5.4.4.5 for the front and back sides of the other specimens of the test fabric.

4.5.4.4.7 Recalibrate periodically, at the minimum between each set of six specimens.

4.5.4.5 Report and evaluation.

4.5.4.5.1 The test method used and any alterations to the method shall be reported.

4.5.4.5.2 Test conditions, including relative humidity, room temperature and conditioning time, shall be stated.

4.5.4.5.3 Equipment names, model numbers and manufacturers shall be listed (see.6.9). Special modifications to equipment shall be described. A description of any in-house built equipment shall be included.

4.5.4.5.4 Identification of materials tested, including name, composition, weave, printed/dyed, finishes, manufacturer and other pertinent information shall be included.

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4.5.4.5.5 Alteration to materials shall be noted together with the test method used, or description of the process applied; e.g., laundering.

4.5.4.5.6 The average time to decay to 10 percent of the maximum voltage shall be included for each of the warp and filling directions, as well as the overall average, for each fabric tested. The maximum voltage level reached by the specimen shall be recorded. It shall be indicated whether the fabric is acceptable or non-acceptable, passing or non-passing.

4.5.4.5.7 Passing the test requires the following: 1) specimen's reaching at least 4000 volts during the 20-second charging period and 2) less than 0.5-second decay time for the charge from maximum voltage level to 10 percent of maximum level.

4.5.5 Liquid repellency test. Place a small specimen of the treated finished camouflage cloth (comprised of all colors in the pattern) on a smooth horizontal surface, face-up. Using a pipette or eye dropper, gently deposit a drop of each liquid specified (see 3.9) on the surface of each shade of the camouflage pattern. After 30 seconds, examine each color of the specimen under light at an angle. Absence of light reflectance at the cloth-drop interface for the n-dode-cane drop shall be taken as evidence of wetting. The test shall be continued for 8 hours for the drops of tri-ethyl phosphate, di-methyl methyl phosphonate, and bis(2 ethyl hexyl) hydrogen phosphite. Absence of light reflectance at the cloth-drop interface for the three liquids after the 8 hour period shall be taken as evidence of wetting. Three specimens taken at various locations across the sample unit shall be tested. Evidence of wetting on any color of one or more specimens shall be considered cause of rejection of the lot represented by the sample.

4.5.6 Toxicity test. When required, (see 6.2), an acute dermal irritation study and a skin sensitization study shall be conducted on laboratory animals. When the results of these studies indicate the finished cloth is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (see 2.3). If the toxicity requirement (see 3.13) can be demonstrated with historical use data, toxicity testing may not be required (see 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The cloth is intended for use in combat camouflage uniforms. Classes 3 and 5 contain a static dissipative fiber in the fiber blend and are not water repellent treated. Classes 2, 4 and 8 are water repellant treated and Classes 1, 6 and 7 are untreated.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of specification.
b. Class of cloth required (see 1.2)
c. The specific issue of individual documents referenced (see 2.2)
d. Woodland or universal camouflage pattern drawing if required (2.2.2).
e. When first article is required (see 3.1 and 4.2).
f. Width of cloth required (see 3.6.2).
g. Length and put up required if other than specified (see 3.14).
h. Inspection conditions (see 4.3.1)
i. When toxicity testing is required (see 4.5.6)
j. Packaging (see 5.1).

6.3 First article. When a first article inspection is required (see 3.1), it will be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Standard sample. For access to information such as patterns, drawings, standard shade samples of cloth etc. for Classes 1-6 go to DLA Troop Support website for their Specification/Pattern/Drawing request form at http://www.dscp.dla.mil/portal/sreqform.aspx. Complete the request form and then submit. For access to samples for Classes 7 and 8 address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Dye combination (all Classes). Both colorfastness and infrared spectral reflectance requirements have been satisfactorily met by the use of both acid and vat dyes.

6.6 Heat setting. The contractors are cautioned to insure that the cloth prior to printing has been properly heat set in order that the final dimensions of the pattern will be within the limits set forth in 3.5.6.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issues, due to the extensiveness of the changes. Classes 3 and 4 were added to this revision due to the addition of the Universal Camouflage pattern. Due to interim changes, Class 3 and Class 5 were changed to contain a static dissipative fiber and Class 6 was added to this revision for use when static dissipation fiber and water repellency are not required. Classes 7 and 8 were added due to the addition of the Operation Enduring Freedom Camouflage Pattern.

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6.8 Static dissipative fiber. Static dissipative fiber also known as P140 is premixed in bale form in a 96% T-420 Nylon /4% P-140 fiber blend and referred to as T420A from the INVISTA, 3 Little Falls Centre, 2801 Centerville Road, Wilmington, DE 19808. The T420A bale is blended with cotton in the yarn formation process.

6.9 Known suppliers of equipment suitable for conducting the Electrostatic Decay Test.

Electro-Tech Systems, Inc. (ETS)
3101 Mt. Carmel Ave
Glenside, PA 19038
http://electrotechsystems.com

Cole-Parmer Instrument
625 East Bunker Court
Vernon Hills, IL 60061-1844
http://www.coleparmer.com

Solomat Corporation
Glenbrook Industrial Park
652 Stamford Road
Stamford, CT 06906

6.10 Operation Enduring Freedom Camouflage Pattern (OCP). Operation Enduring Freedom Camouflage Pattern (OCP) has previously been referred to as MultiCam® Camouflage Pattern.

6.11 Subject term (key word) listing

Cover, Helmet
Coveralls, Mechanic
Operation Enduring Freedom - OCP
Static Dissipative Fiber
Water repellent treated
Woodland
Universal

Custodian:
Army – GL
Navy – NU
Air Force – 11

Preparing activity:
DLA – CT

Review activities:
Army – MD
Navy – MC

Project Number: 8305-2010-002

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